Table of Contents

iMotions Lab > Getting Started

Analysis Overview 2

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Analysis Overview

Overview of data analysis in iMotions software.

Summary

- Recorded data can be reviewed in any Replay and Analysis Window.
    - Respondent Replays are available immediately after data collection.
- Create an Analysis to setup and access:
    - Event Markers (time points) and Annotations (time periods)
    - Aggregate Results
    - Signal Processing (R Notebooks)
    - Eye tracking analysis (AOI, Gaze Mapping, Heatmaps)
    - Additional information and Help Center article links provided below

Creating an Analysis

Create a New Analysis

1. Select a study that contains at least one respondent recording.
2. Hover over the study name Library and click the Add Analysis button that appears.

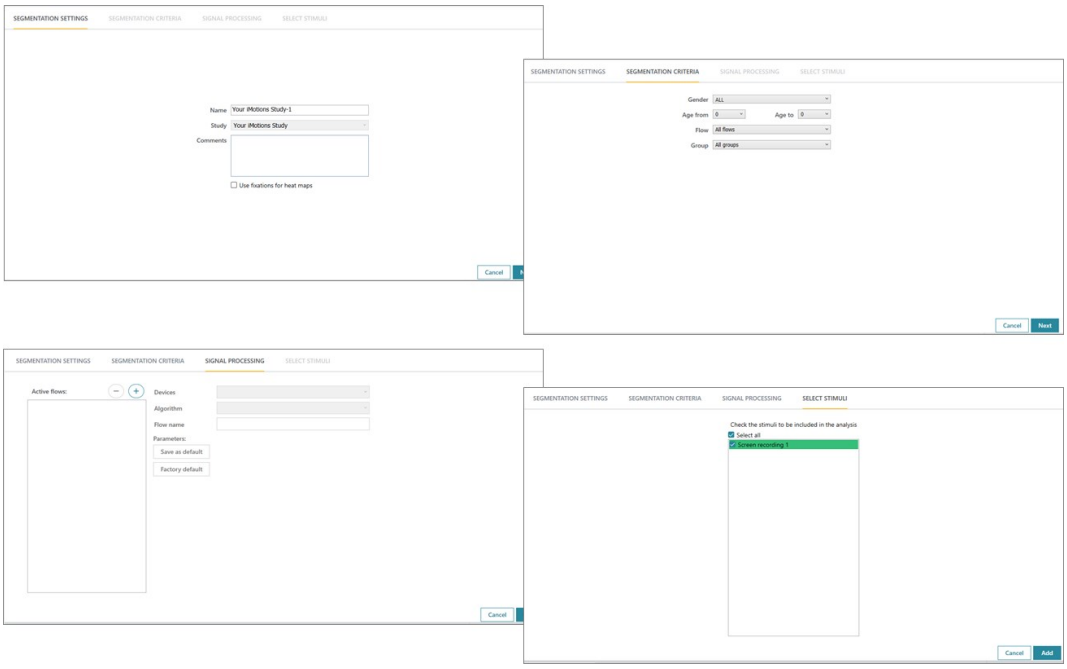


Set Up Analysis

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Note: Your Analysis can be edited at any time.

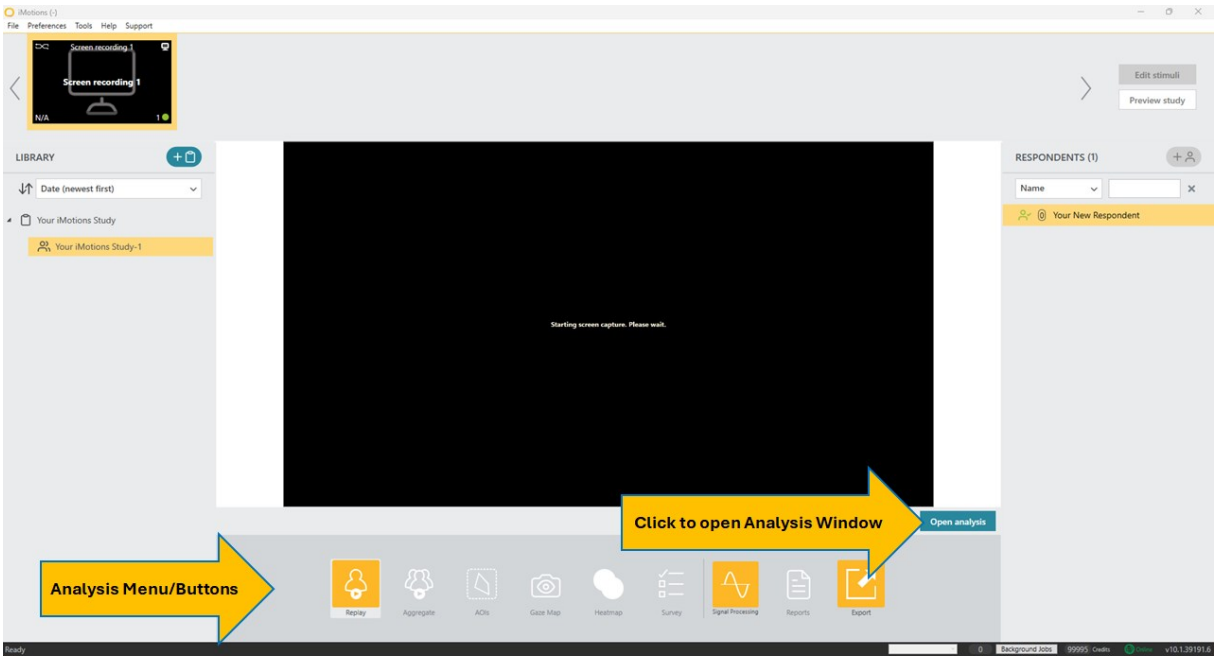
1. Enter a Name and Comments (as needed) then click Next.
2. Define Segmentation Criteria (as needed) and click Next.
    - Segmentation Criteria allow you to select respondents to include in the analysis.
    - This can be done manually (by checking/unchecking in the Respondent list) or automatically (by Gender, Age, Flow, and Group).
3. Set up Signal Processing (as needed) and click Next.
    - Active flows - List of Signal Processing Algorithms in the analysis
    - Devices - Options depend on the sensors/data in the study
    - Algorithm - Options depend on the selected device
4. Select Stimuli (as needed).
    - All stimuli are selected by default.
    - To facilitate analysis, select stimuli that are relevant for analysis.
    - At least one stimulus must be selected.
5. Click Add to create the analysis
    - The new analysis will now appear in the drop-down list below the study
    - Analysis menu buttons will also appear at the bottom of the main iMotions window (according to data and stimuli in the analysis)
    - Only the selected respondents and stimuli will be shown/available in the study.



iMotions Analysis settings

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iMotions Analysis



General Tools

- Replay and Analysis Windows are used to review recorded data.
- Event Markers (time points) and Annotations (time periods) are created, defined, and edited in these windows.
- Results are calculated over Annotations, according to the Signal Processing and/or AOIs included in your analysis.
- These results are available in Replay Windows and Exports.

Signal Processing (R Notebooks)

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- What are R Notebooks? - R Notebooks are Signal Processing scripts used for sensor data analysis
- Signal Processing (R Notebooks) are available for analyzing data from specific sensors.
    - Eye tracking Blink Detection
    - Facial Expression Analysis Thresholding and Aggregation
    - Voice Analysis
    - GSR Peak Detection and Signal Quality
    - EEG Power Spectral Density Inter Subject Correlation Frontal Asymmetry
    - Respiration Processing and Moment-to-Moment Aggregation
    - ECG Heart Rate and HRV
    - EMG Processing
    - Skin Temperature
    - SpO2 Oxygen Saturation

Eye Tracking Analysis

- Eye tracking analysis typically involves defining Areas of Interest (AOIs) - areas within a visual stimulus/recording for calculating AOI Metrics (in the AOI replay window).
- AOIs are created/edited in the AOI Replay Window using the AOI Editor or Automated AOI tools.
- Eye tracking results are available as Heat Maps and Scan Paths for 'static' stimuli in any iMotions analysis with eye tracking data.
- Gaze Map Analysis can facilitate AOI analysis of 'dynamic' stimuli (such as Screen/Website Recordings or Eye Tracking Glasses camera recordings, controlled by respondents) by mapping eye tracking data onto user-defined, reference images. This is necessary for aggregate results, such as Heatmaps and Scan Paths.

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